

Principles of Inheritance and Variation

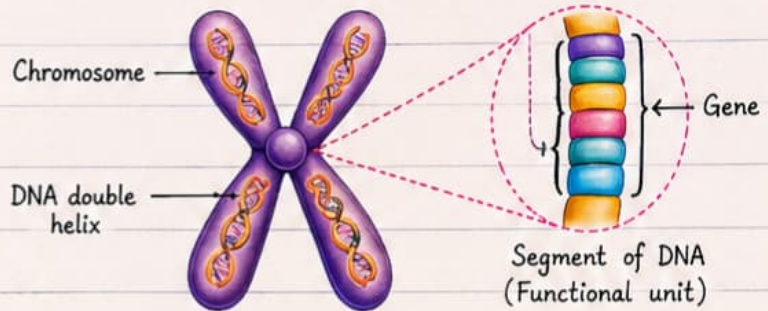


“Inheritance is the passage of information from one generation to the next, while variation creates diversity in life.”

Key Concepts

- Mendelian Principles of Inheritance
- Chromosomal Theory of Inheritance
- Linkage and Recombination
- Sex Determination
- Variation - Types and Causes

Chromosome & Gene



1. Mendel's Work

- Gregor Mendel is known as the “Father of Genetics”.
- He conducted experiments on pea plants (*Pisum sativum*).
- He selected 7 contrasting characters.
- His work led to the formulation of Mendel's Laws.

Seven Contrasting Characters in Pea Plant

Character	Contrasting Traits	
• Seed shape	Round / Wrinkled	 
• Seed colour	Yellow / Green	 
• Flower colour	Violet / White	 
• Pod shape	Inflated / Constricted	 
• Pod colour	Green / Yellow	 
• Flower position	Axial / Terminal	 
• Plant height	Tall / Dwarf	 

2. Mendel's Laws of Inheritance

★ **Law of Dominance**: When two contrasting traits are brought together in a hybrid, only one (dominant) is expressed, while the other (recessive) is suppressed.

Example: Tall (T) is dominant over dwarf (t) in pea plants.

★ **Law of Segregation**: The two alleles of a gene separate during gamete formation, and only one allele goes into a gamete.

Example: In Tt plant, gametes will be T and t in equal proportion.

Key Point

Traits are controlled by genes.
Genes occur in pairs (alleles).
One allele from each parent is inherited.

DNA - The Genetic Material



- DNA is a double helix molecule.
- It is made up of nucleotides.
- Gene is a segment of DNA that codes for a particular protein.
- Genes are located on chromosomes.

Nucleotide Components

- 1) Nitrogenous base (A, T, G, C)
- 2) Pentose sugar (Deoxyribose)
- 3) Phosphate group



3. Mendel's Experiments on Pea Plant

- Mendel selected pea plant because it has many **contrasting characters**.
- Pea plant shows **true-breeding** nature.
- Flowers are **bisexual** and naturally **self-pollinated**.
- Mendel performed **artificial cross-pollination** to control pollination.

4. Mendel's Monohybrid Cross

- Cross between two plants differing in **one character**.
- Example**: Seed shape (**Round vs Wrinkled**)
- Round (**R**) is **dominant** over wrinkled (**r**).
- Wrinkled character is **recessive**.

Contrasting Characters in Pea Plant

Seed Shape	 Round	 Wrinkled
Seed Colour	 Yellow	 Green
Flower Colour	 Violet	 White
Pod Shape	 Inflated	 Constricted
Pod Colour	 Green	 Yellow
Flower Position	 Axial	 Terminal
Plant Height	 Tall	 Dwarf

Punnett Square for Monohybrid Cross

$$Rr \times Rr$$

Gametes R , r R , r

♀ \ ♂	R	r
R	RR (Round)	Rr (Round)
r	Rr (Round)	rr (Wrinkled)

Genotypic Ratio = 1 RR : 2 Rr : 1 rr

Phenotypic Ratio = 3 Round : 1 Wrinkled

Monohybrid Cross (Example : Seed Shape)

P Generation

Round (RR)

×

Wrinkled (rr)



F₁ Generation

All Round (Rr)



F₁ Selfing

(Rr × Rr)

F₂ Generation

Round



Wrinkled



3

1

Ratio = 3 : 1

5. Laws of Inheritance (Mendel's Laws)

1) Law of Dominance

- When two contrasting traits are present together, only the **dominant trait** is expressed.
- Recessive** trait remains suppressed.



2) Law of Segregation

- The two alleles of a gene separate during gamete formation.
- Each gamete receives only **one allele**.

3) Law of Independent Assortment

- Alleles of different genes assort **independently** during gamete formation.
- This law holds true for dihybrid or polyhybrid crosses.



Did You Know ?

Mendel is called "Father of Genetics".

Important Note

- Genes** control characters.
- Traits** are inherited from parents.
- Variation** makes every individual unique.
- Mendel's laws are the foundation of **genetics**.

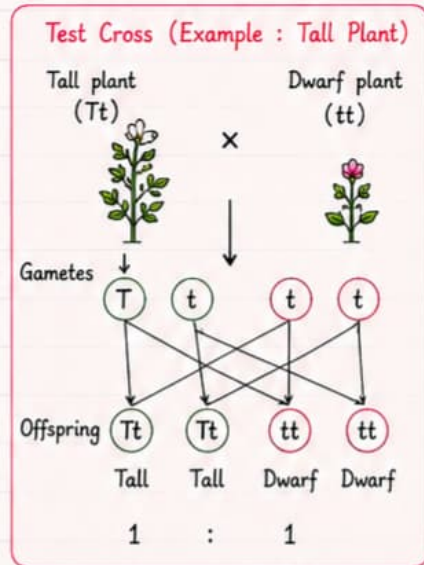
Quick Revision

- Gene → Unit of inheritance
- Allele → Different forms of gene
- Dominant → Expressed trait
- Recessive → Suppressed trait
- Variation → Differences among individuals

Inheritance and Variation

6. Test Cross

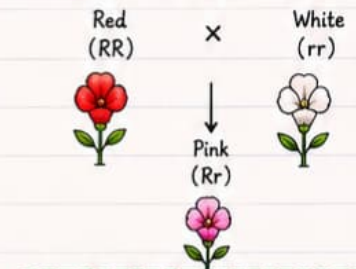
- It is a cross between the F_1 hybrid and the **recessive parent**.
- It is used to determine the **genotype** of the dominant phenotype.
- If all offspring show dominant trait $\rightarrow F_1$ was **homozygous dominant**.
- If offspring show **1:1 ratio** $\rightarrow F_1$ was **heterozygous**.



7. Incomplete Dominance

- In this case, **neither** allele is completely dominant.
- The heterozygote shows an **intermediate phenotype**.

Example : Snapdragon flower



Cross : $RR \times rr \rightarrow$ all Rr (Pink)

Selfing : $Rr \times Rr \rightarrow$

RR (Red) : Rr (Pink) : rr (White)
= 1 : 2 : 1

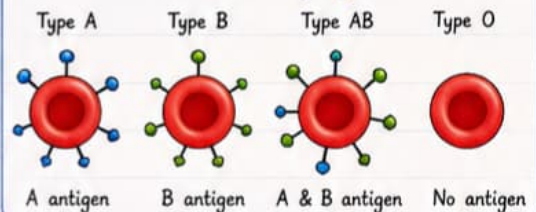
8. Codominance

- Both alleles are **fully expressed** in the **heterozygote**.
- Both the characters appear **together**.

Example : Human ABO blood group

Genotype	Phenotype	Blood Group
$I^A I^A$	Only A antigen	A
$I^B I^B$	Only B antigen	B
$I^A I^B$	Both A and B antigen	AB
ii	No antigen	O

ABO Blood Group System



9. Multiple Alleles

- More than two **alleles** control a single character.
- Example :** ABO blood group (I^A , I^B and i)
- Only **two alleles** are expressed in a genotype.

Hierarchy of Dominance in ABO System

$$I^A = I^B > i$$

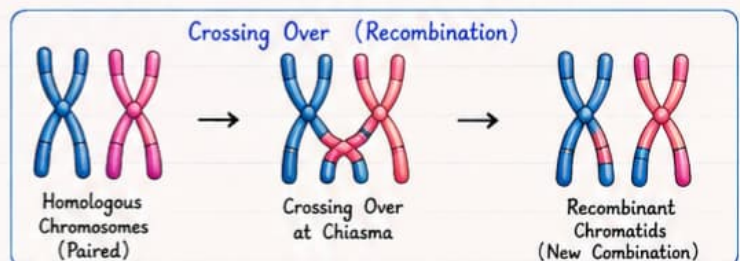
I^A and I^B are **codominant** with each other but both are **dominant** over i .

Possible Genotypes and Phenotypes

Blood Group	Genotype	Phenotype
A	$I^A I^A$, $I^A i$	A
B	$I^B I^B$, $I^B i$	B
AB	$I^A I^B$	AB
O	ii	O

10. Linkage and Recombination

- Genes located on the same chromosome tend to be inherited together and are called **linked genes**.
- Genes on different chromosomes assort **independently**.
- Crossing over** during prophase-I of meiosis causes exchange of segments $\rightarrow$ **recombination**.



Key Takeaways

- ✓ Mendel's laws explain inheritance of traits.
- ✓ Genes are the units of inheritance.
- ✓ **Variation** arises due to **mutation**, recombination and independent assortment.
- ✓ These principles form the base of **genetics**.

Important Terms

- Gene** : Unit of heredity
- Allele** : Alternative form of a gene
- Trait** : A specific character
- Dominant** : Expressed trait
- Recessive** : Suppressed trait
- Variation** : Differences among individuals

Remember

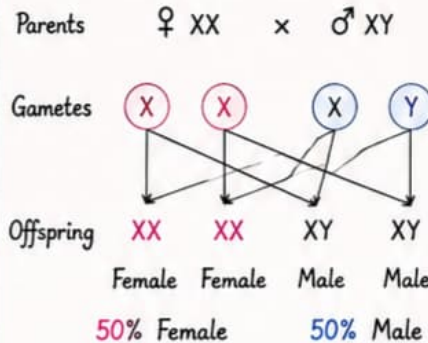
"Inheritance creates similarity, variation creates diversity."

Inheritance and Variation

11. Sex Determination

- In humans, sex is determined by **sex chromosomes**.
- Female → **XX (Homogametic)**
- Male → **XY (Heterogametic)**
- Ovum always carries **X** chromosome.
- Sperm carries either **X** or **Y** chromosome in equal proportion.
- Male decides the sex of the child.

Sex Determination in Humans



Key Point ★

In human beings, females are homogametic (XX) and males are heterogametic (XY). Thus, males produce two types of sperms (X-bearing and Y-bearing) in equal number.

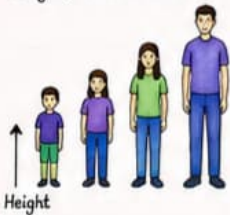
12. Variation – Types and Causes

Types of Variation

Continuous Variation

Small differences which occur gradually.

Example: Height, weight, skin colour.



Discontinuous Variation

Distinct categories with no intermediates.

Example: Blood groups, earlobe type, tongue rolling.



Causes of Variation

- Mutation** – sudden changes in DNA.
- Recombination** – new gene combinations during meiosis.
- Independent assortment** – random distribution of chromosomes.
- Gene flow** – movement of genes from one population to another.
- Environmental factors** – nutrition, climate, drugs, lifestyle etc.

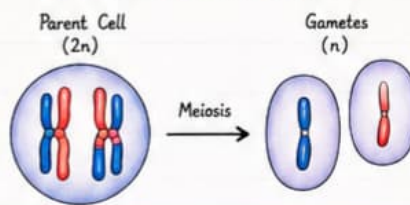
Examples

- Height
- Skin colour
- Blood groups
- Ear lobe
- Tongue rolling

13. Chromosomal Theory of Inheritance

- Proposed by **Sutton and Boveri**.
- Genes are located on chromosomes in a linear order.
- Chromosomes act as carriers of genes.
- During gamete formation, segregation of chromosomes leads to segregation of genes.

Chromosome Behaviour in Meiosis



Remember 💡

The behaviour of chromosomes during meiosis explains Mendel's principles of inheritance.

14. Importance of Variation

- Provides raw material for evolution.
- Helps organisms to adapt to changing environment.
- Increases chances of survival of a species.



15. Significance of Mendel's Work

- ✓ Discovered fundamental laws of inheritance.
- ✓ Used simple organisms (pea plant) for experiments.
- ✓ Gave the concept of dominant and recessive traits.
- ✓ Laid the foundation of modern genetics.
- ✓ His work opened a new era in biology.



Quick Revision ★

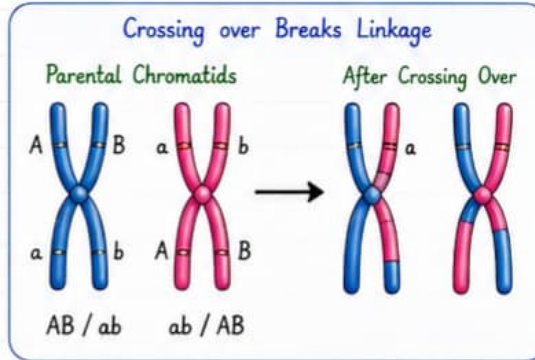
- ★ Genes are units of inheritance.
- ★ Traits are controlled by genes.
- ★ Inheritance → Similarity
- ★ Variation → Diversity
- ★ Mendel's laws → Dominance, Segregation, Independent Assortment.
- ★ Chromosomes are carriers of genes.

Summary

Inheritance ensures continuity of life by passing traits from parents to offspring, while variation brings diversity and helps species survive and evolve.

16. Linkage and Recombination

- Genes located on the same chromosome tend to be inherited together and are called **linked genes**.
- Linkage does not allow independent assortment of genes.
- Crossing over** during prophase-I of meiosis may break linkage and new combinations are formed.



Key Point ★

Crossing over increases genetic variation and breaks the linkage between genes.

17. Mendelian Disorders

Some genetic disorders are caused by **single gene** mutations and follow Mendelian inheritance.

Remember

Most genetic disorders are rare in population but important for genetic counselling.

Disorder	Gene / Cause	Type	Symptoms
Haemophilia	Mutation in F8 gene	X-linked recessive	Failure of blood to clot
Colour Blindness	Mutation in opsin gene	X-linked recessive	Inability to distinguish certain colours
Sickle Cell Anaemia	Mutation in HBB gene	Autosomal recessive	Abnormal haemoglobin, sickle shaped RBCs
Phenylketonuria (PKU)	Mutation in PAH gene	Autosomal recessive	Mental retardation if not treated early
Thalassemia	Mutation in globin genes	Autosomal recessive	Anaemia, weakness

18. Gene Interaction - Epistasis

In some cases, interaction between genes produces a new type of **phenotypic ratio**.

Example : Coat colour in dogs

Genotype	Phenotype	Ratio	Explanation
E_B_	Black	9	Gene E allows colour expression, B gives black
E_bb	Brown	3	Gene E allows colour, b gives brown
eeB_	Yellow	3	Gene e gives yellow (B is masked)
eebb	White	1	No pigment produced

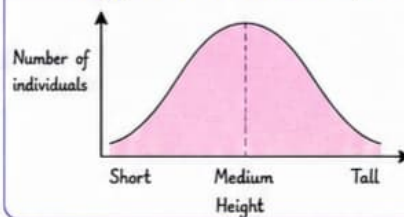
Other Types of Interaction

- Complementary genes - both genes necessary.
- Duplicate genes - either one gene is sufficient.
- Inhibitory genes - one gene suppresses another gene.

19. Polygenic Inheritance

- Many characters are controlled by two or more pairs of genes called **polygenes**.
- These characters show continuous variation.
- Example** : Height, skin colour, eye colour in humans.

Polygenic Inheritance - Example : Height in Humans



- Many genes with small additive effects control the character.
- Produces a **bell-shaped frequency distribution curve**.
- No clear cut classes.

Chapter Summary

- ☑ Inheritance is transmission of characters from parents to offspring.
- ☑ Mendel's laws explain inheritance of traits.
- ☑ Variation arises due to mutation, recombination and independent assortment.
- ☑ Linkage and crossing over produce new gene combinations.
- ☑ Sex determination is controlled by sex chromosomes.
- ☑ Polygenic inheritance shows continuous variation in traits.

Important Formulae

- Test Cross for Dominant Trait**
 $Tt \times tt \rightarrow 1 \text{ Tall} : 1 \text{ Dwarf}$
- Monohybrid Cross**
 $Rr \times Rr \rightarrow 1 RR : 2 Rr : 1 rr$
- Dihybrid Cross**
 $RrYy \times RrYy \rightarrow 9 : 3 : 3 : 1$
- Genotypic Ratio ($Rr \times Rr$)**
 $1 RR : 2 Rr : 1 rr$
- Phenotypic Ratio ($Rr \times Rr$)**
 $3 \text{ Round} : 1 \text{ Wrinkled}$

NEET Exam Tip

- Understand Mendel's laws thoroughly.
- Practice crosses and ratios.
- Learn common genetic disorders and their inheritance patterns.
- Focus on linkage, crossing over and polygenic inheritance.
- Diagrams and tables help in quick revision.